

ADIL ARYA

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EDUCATION

Columbia University

Master of Science in AI, Concentration in Robotics and Perception

• Incoming Student

New York City, New York

Sep 2026 – Dec 2027

University of Minnesota - Twin Cities

Bachelor of Science in Computer Science

• **Honors:** 3.912 / 4.0 GPA, 5x CSE Dean's List, High Honors

• **Courses:** Algorithms & Data Structures, Operating Systems, Machine Learning, Computer Vision, Robotics, NLP, Artificial Intelligence I & II

Minneapolis, Minnesota

Sep 2024 – May 2026

WORK EXPERIENCE

Grit Financial Inc.

Software Engineer Intern

• Joined the core engineering team building Grit's fintech platform, ramping on the production codebase and contributing to ongoing feature development.

Remote

Feb 2026 – Present

Minnesota Robotics Institute

Undergraduate Researcher

• Benchmarked GLOMAP against COLMAP for 3D reconstruction across 30 trials on Chamfer distance, precision/recall, and F-score; GLOMAP raised F-score to **0.89 (from 0.79)** and cut Chamfer error **~30%**, leading the lab to adopt it as the reconstruction backbone (Python, NumPy, Pandas).

• Built **CableMind**, an autonomous cable-routing system on a UFactory Lite6 arm: an AprilTag-driven modular state machine (pose estimation → motion planning → xArm SDK execution) achieving **100% success on single-clip routing** and 60% on randomized 4-clip sequences, plus a LoRA-fine-tuned OpenVLA (Llama-2-7B) policy mapping language instructions to end-effector trajectories.

Minneapolis, Minnesota

Oct 2024 – May 2026

Divertica Inc.

Computer Vision Intern

• Built an end-to-end CV pipeline converting **4K/60fps** in-store video into structured retail planograms via frame stitching, object detection, and OCR.

• Benchmarked **YOLOv8s** and **YOLOv12s** (fine-tuned on **SKU110K**), **ResNet50**, and **ImageNet**-pretrained OCR, reaching **0.87 F1 / 0.90 mAP50** on shelf detection; found YOLOv12 offered no meaningful gain over YOLOv8 and surfaced a classification bottleneck at **~630K SKU classes**, proposing a retrieval-based alternative.

Remote

Jun 2025 – Aug 2025

PROJECTS

TrialFind

Origin House O1 Summit Hackathon

• **1st Place, TinyFish Sponsor Track** (24-hour hackathon, team of 3): agentic system matching cancer patients to clinical trials via a **Featherless** summarization agent and parallel **TinyFish** browser agents across trial registries.

• Pooled sponsor credits to scale from **2 to 6 parallel browser agents**, enabling real-time multi-site search.

Apr 2026 – Apr 2026

Recursive Ray Tracer

CSCI 5607, University of Minnesota

• Built a recursive ray tracer from scratch in **C++** with ray intersection for spheres, triangle meshes, cylinders, cones, and ellipsoids, plus barycentric smooth shading and UV-mapped textures.

• Implemented **Blinn-Phong** illumination, reflection/refraction via **Schlick's approximation** with total internal reflection, **Beer's-law** attenuation, and **Monte Carlo** diffuse interreflection.

Jan 2026 – Apr 2026

RESEARCH AND PUBLICATIONS

Imitation of Imitations: Model Collapse under Synthetic Training Data

Published on *arXiv preprint (in preparation)*

Jun 2026

SKILLS

Programming Languages: Python, C++, SQL, JavaScript / TypeScript, MATLAB

Web & Backend: React, Next.js, Node.js, FastAPI, Flask

Tools & Infrastructure: Git, Docker, Linux, Jupyter, Colab

ML & Deep Learning: PyTorch, JAX, Hugging Face Transformers, LoRA fine-tuning, scikit-learn, NumPy, Pandas, CNN / ResNet / MLP

Computer Vision: OpenCV, torchvision, PIL, Ultralytics YOLO, Structure-from-Motion (COLMAP, GLOMAP)

NLP & LLMs: LangChain, OpenAI and Anthropic APIs, Continued pretraining and synthetic data generation